Milad Mohammadi

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Highlights

- Published at **EACL 2026** as second author; first-author preprint under review at **NAACL 2027**.
- Master's GPA: **19.14/20**; ranked **1st nationwide** in Iran's IT Engineering master's entrance exam.
- **3+ years** of industry experience in AI engineering, including production LLM agents and RAG systems.

Research Interests

Large Language Models (LLMs); Multimodal Large Language Models (MLLMs); Language Agents; Self-Evolving Language Agents; Modeling Human Behavior with LLMs; Personalization & Recommender Systems via LLMs; User Profiling; Agent-based Simulation; LLM Safety; Privacy-Preserving Foundation Models

Education

University of Tehran Sep. 2023 – Sep. 2026

M.Sc. in IT Engineering | GPA: **19.14/20** | Thesis grade: **Excellent** (highest grade)

- **Thesis:** *Design and Implementation of an Intelligent Assistant for Clinical Assessment and Diagnosis of Mental Disorders Using Large Language Models*.

University of Isfahan Sep. 2018 – Jul. 2023

B.Sc. in Computer Engineering (IT major) | Major GPA: 15.72/20

- **Final Project:** *Design and Development of a Structured Feedback Collection and Evaluation System for University Teaching and Research Assistants to Enhance Instructional and Research Support Quality*.

Publications

From Dialogue to Diagnostic Hypotheses: Evaluating LLM Agents for Psychiatric Assessment

First author *arXiv preprint (2026); under review at NAACL 2027*

SchemaGraphSQL: Efficient Schema Linking with Pathfinding Graph Algorithms for Text-to-SQL on Large-Scale Databases

Second author *Findings of EACL 2026*

UT-NLP at SemEval-2025 Task 11: Evaluating Zero-Shot Capability of GPT-4o mini on Emotion Recognition via Role-Play and Contrastive Judging

Co-first author (equal contribution) *SemEval 2025*

Research Experience

NLP Lab, University of Tehran Jan. 2024 – Sep. 2026

Research Assistant (M.Sc.) | Advisor: Prof. Heshaam Faili

- Contributed to lab research through weekly discussions, paper critiques, project planning, and collaborative development.
- Mentored incoming M.Sc. students in defining and scoping research problems involving language agents.

Computational Cognitive Psychology Lab, University of Tehran Jan. 2024 – Sep. 2026

Research Assistant (M.Sc.) | Advisor: Prof. Hadi Moradi

- Initiated a research track on digital mental health applications of large language and vision-language models.

Honors and Awards

Ranked 1st Nationwide in IT Engineering, Master's Entrance Exam (Konkur), 2023

National examination for admission to IT engineering master's programs in Iran; ranked first among 4,696 candidates.

Industry Experience

IntRest (Canada, Remote) Feb. 2026 – Present

Senior AI Engineer (Part-time)

- Lead the architecture and end-to-end delivery of 7+ AI food-analysis capabilities; design a **modular Python platform** with separate API, domain, and food-data layers and 60+ typed operations.
- Architect **grounded LLM-agent workflows** combining hybrid retrieval, validated tool evidence, and deterministic nutrient calculations over canonical USDA and CNF food records.
- Engineer **production reliability** with asynchronous concurrency controls, deadlines, retries, and layered caching; deliver typed API contracts, regression tests, Langfuse tracing, and frontend/backend integration.

Adan AI (Tehran, On-site) Jun. 2024 – Nov. 2025

AI Software Engineer (Part-time)

- Led the architecture and delivery of an agentic customer-service chatbot for a national bank: translated product requirements into user stories; designed a ReAct/Reflexion workflow with over 15 tool steps; implemented and reviewed LangGraph and OpenAI modules; defined and tracked production evaluation metrics.
- Built a Persian PDF pipeline with table- and chart-aware parsing, chunking, and entity extraction; cut processing costs by 50% and improved retrieval accuracy by 8 percentage points on a human-annotated evaluation set.
- Developed an automated FAQ chatbot generator that cleans Excel FAQs, expands seed data, and deploys chatbots without manual coding, reducing onboarding time from days to under one hour.

Teaching Experience

University of Tehran | School of Electrical and Computer Engineering

Co-Chief Teaching Assistant

- **Natural Language Processing** Fall 2025; Spring 2026
  - Co-led teaching assistant teams for two NLP course offerings, including a 12-member team; reviewed the design and content of TA-prepared programming assignments and quizzes.
  - Coordinated assignments, quizzes, and paper presentations, including schedules and logistics; graded midterm and final exams.
  - Served as students' primary point of contact for assignment guidance and course-related questions.

Teaching Assistant

- **Large Language Models** Spring 2025
- **Natural Language Processing** Fall 2024
- **Intelligent Information Retrieval** Fall 2024
- **Deep Learning & Neural Networks** Fall 2024
  - Designed assignments and final projects, including Text-to-SQL Framework, ChatFood, NanoSparseRetriever, and an LLM fine-tuning project; developed starter code and grading rubrics.
  - Delivered hands-on workshops on language agents, RAG systems, and LLM architecture and fine-tuning. Selected recordings: YouTube playlist (8 videos; 1,900+ combined views).
  - Graded assignments and quizzes; provided individual guidance on course concepts and programming assignments.

Technical Skills

Software Engineering	Python; modular design; automated testing (pytest); packaging (uv, pip-tools)
Agent Engineering	OpenAI Agents SDK; Tool-Use/Planning; multi-agent systems; agent memory
Retrieval & RAG	Milvus; LanceDB; hybrid search; reranking; document parsing; chunking
LLM Fine-tuning	Hugging Face Transformers; PEFT (LoRA, QLoRA); TRL; tokenization
Evaluation & Monitoring	Retrieval evaluation; LLM-as-judge; Langfuse; metrics, logs, and traces
Backend & Databases	FastAPI; Pydantic; async I/O; PostgreSQL; SQLite
Machine Learning	PyTorch; scikit-learn; NumPy; pandas; Matplotlib; feature engineering
APIs & Authentication	REST; gRPC; WebSockets; OAuth2/OpenID Connect; JWT
Deployment & MLOps	Model serving (vLLM); Docker; Docker Compose; CI/CD
Systems & Delivery	Requirements analysis; software architecture; code review; technical documentation

English Proficiency

TOEFL iBT (Test Center) July 25, 2026

Overall: **5.5/6** | ETS comparable score: **110/120** | CEFR C1

Reading: **6.0**/6 Listening: **6.0**/6 Writing: **5.5**/6 Speaking: **5.0**/6

Selected Projects

Text-to-Insight Data Analyst Agent GitHub

LLM course final project | Team of 3

- Co-developed an agent that queries BIRD databases in SQLite, generates Python analyses and plots, and produces reports from natural-language questions; evaluated SQL and report accuracy using an LLM-as-judge framework.

MedYar: Medical Appointment Agent GitHub

48-hour hackathon | Team of 3

- Co-developed a ReAct-style assistant that collects symptoms and preferences, recommends medical specialties, checks availability, and books appointments on *Paziresh24* through web automation.

Leadership and Outreach

ACM Student Chapter, University of Isfahan

Committee Member

- Organized programming contests and taught algorithm design; participated in weekly ICPC training sessions.

TEDxUniversityOfIsfahan

Team Lead & Organizer

- Founded *TEDxUniversityOfIsfahan* in 2019 and led event planning, budgeting, sponsorship, marketing, and speaker coaching; the student community continues the annual series, with talks attracting thousands of views.

References

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